

$$3. \vec{H}(\rho, t) = \frac{H_0}{\sqrt{\rho}} \cos \left[\omega t - \beta \rho + \frac{\pi}{2} - \frac{\pi}{2} \right] \vec{a}_\phi$$

$$\vec{H}(\rho, t) = \frac{H_0}{\sqrt{\rho}} \cos [\omega t - \beta \rho] \vec{a}_\phi$$

$$\vec{H}(\rho) = \frac{H_0}{\sqrt{\rho}} e^{-j\beta\rho} \vec{a}_\phi$$

Example (9):

Re-write (in time-domain) the following equations:

$$1. \vec{E}(x, y) = j e^{j\beta x} \vec{a}_z + j e^{j\beta y} \vec{a}_z$$

$$2. \vec{H}(r) = j \frac{H_0}{r} e^{-j2\pi r} e^{j\phi} \vec{a}_\phi$$

Solution:

$$1. \vec{E}(x, y) = [j e^{j\beta x} + j e^{j\beta y}] \vec{a}_z$$

$$\vec{E}(x, y, t) = \text{Re} \{ j e^{j\beta x} + j e^{j\beta y} \} \vec{a}_z$$

$$\vec{E}(x, y, t) = \text{Re} \left\{ e^{j\frac{\pi}{2}} e^{j\beta x} + e^{j\frac{\pi}{2}} e^{j\beta y} \right\} \vec{a}_z$$

$$\vec{E}(x, y, t) = \left(\cos \left[\omega t + \beta x + \frac{\pi}{2} \right] + \cos \left[\omega t + \beta y + \frac{\pi}{2} \right] \right) \vec{a}_z$$

$$2. \vec{H}(r) = j \frac{H_0}{r} e^{-j2\pi r} e^{j\phi} \vec{a}_\phi$$

$$\vec{H}(r, t) = \frac{H_0}{r} \cos \left[\omega t - 2\pi r - \frac{\pi}{2} + \phi \right] \vec{a}_\phi$$

$$\begin{aligned} j &= e^{j\frac{\pi}{2}} \\ -j &= e^{-j\frac{\pi}{2}} \end{aligned}$$

Example (10):

A Magnetic field intensity is given by:

$$\vec{H}(\rho) = j \frac{H_0}{\rho} e^{-j\beta\rho} \vec{a}_\phi$$

Find:

1. $\vec{H}^*$

2. $\vec{H}(\rho, t)$

3. $\vec{H}(x, y)$

4. $\vec{H}(x, y, t)$

5. $\vec{D}(\rho)$, where $\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$

6. $\vec{P}_v$ where $\vec{P}_v = \nabla \cdot \vec{D}$

7. $\nabla^2 \vec{H}_\phi$

Solution:

1. $\vec{H}(\rho) = j \frac{H_0}{\rho} e^{-j\beta\rho} \vec{a}_\phi$

$$\vec{H}^*(\rho) = (-j) \frac{H_0}{\rho} e^{+j\beta\rho} \vec{a}_\phi$$

2. $\vec{H}(\rho, t) = \vec{a}_\phi \frac{H_0}{\rho} \text{Re} \{ j e^{-j\beta\rho} \}$

$$= \vec{a}_\phi \frac{H_0}{\rho} \text{Re} \left\{ e^{j\frac{\pi}{2}} e^{-j\beta\rho} \right\}$$

$$= \vec{a}_\phi \frac{H_0}{\rho} \text{Re} \left\{ e^{j[-\beta\rho + \frac{\pi}{2}]} \right\}$$

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• لا يجب دمج conjugate أي عدد
فما علينا إلا أن نغير إشارة
الدور فكل سبيل المثال

$$\vec{E} = j e^{-j\beta x} e^{-j\omega t} \vec{a}_z$$

$$\vec{E}^* = (-j) e^{+j\beta x} e^{-j\omega t} \vec{a}_z$$

• تذكر أن

$$\vec{H}(\rho, t) = \boxed{} \text{Re} \{ \boxed{} \}$$

كل شيء
real

كل شيء
imaginary
جوي

$$\vec{H}(\rho, t) = \vec{a}_\phi \frac{H_0}{\rho} \cos \left[\omega t - \beta \rho + \frac{\pi}{2} \right]$$

3.

$$\begin{aligned} \tilde{H}_x &= \tilde{\vec{H}} \cdot \vec{a}_x \quad \left. \begin{array}{l} \text{من الجدول} = -\sin \phi \\ \text{ابحار المركبة} \\ \text{باتجاه } x \end{array} \right\} \\ &= j \frac{H_0}{\rho} e^{-j\beta\rho} [\vec{a}_\phi \cdot \vec{a}_x] \\ &= j \frac{H_0}{\rho} e^{-j\beta\rho} [-\sin \phi] \\ &= -j \frac{H_0}{\sqrt{x^2+y^2}} \left(\frac{y}{\sqrt{x^2+y^2}} \right) e^{-j\beta\sqrt{x^2+y^2}} \quad \left. \begin{array}{l} \text{تحويل} \\ \text{المتغيرات} \\ (r, \phi) \\ \text{الى} \\ (x, y) \end{array} \right\} \\ &= \frac{-j H_0 y}{(x^2+y^2)} e^{-j\beta\sqrt{x^2+y^2}} \end{aligned}$$

$$\begin{aligned} \tilde{H}_y &= \tilde{\vec{H}} \cdot \vec{a}_y \\ &= j \frac{H_0}{\rho} e^{-j\beta\rho} [\vec{a}_\phi \cdot \vec{a}_y] \\ &= j \frac{H_0}{\rho} e^{-j\beta\rho} \cos \phi \\ &= j \frac{H_0}{\sqrt{x^2+y^2}} e^{-j\beta\sqrt{x^2+y^2}} \frac{x}{\sqrt{x^2+y^2}} \\ &= \frac{j H_0 x}{(x^2+y^2)} e^{-j\beta\sqrt{x^2+y^2}} \end{aligned}$$

$$\tilde{H}_z = \tilde{\vec{H}} \cdot \vec{a}_z = j \frac{H_0}{\rho} e^{-j\beta\rho} [\vec{a}_\phi \cdot \vec{a}_z] = 0$$

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- للتحويل من نظام إحداثي

إلى نظام إحداثي ثاني يجب أن نعرف طين صما:

1. تحويل المركبات حسب الجدول

2. تحويل المتغيرة حسب العلاقات

- في المطلب الثالث قاتنا

سوف تحول النظام من

الأحداثيات الأسطوانية إلى الأحداثيات الكارتيزية

- تذكر أن

$$\rho = \sqrt{x^2+y^2}$$

$$\phi = \tan^{-1} \frac{y}{x}$$

$$\begin{aligned} \sin \phi &= \frac{y}{\rho} \\ &= \frac{y}{\sqrt{x^2+y^2}} \end{aligned}$$

$$\cos \phi = \frac{x}{\sqrt{x^2+y^2}}$$

$$\therefore \vec{H}(x, y) = j \frac{H_0}{(x^2 + y^2)} e^{-j\beta\sqrt{x^2 + y^2}} [-y \vec{a}_x + x \vec{a}_y]$$

$$4. \quad \vec{H}(x, y, t) = \frac{H_0}{x^2 + y^2} [-y \vec{a}_x + x \vec{a}_y] \operatorname{Re} \left\{ j e^{j\frac{\pi}{2}} e^{-j\beta\sqrt{x^2 + y^2}} e^{j\omega t} \right\}$$

$$= \frac{H_0}{(x^2 + y^2)} [-y \vec{a}_x + x \vec{a}_y] \operatorname{Re} \left\{ e^{-j \left[\beta\sqrt{x^2 + y^2} - \frac{\pi}{2} \right]} e^{j\omega t} \right\}$$

$$= \frac{H_0}{(x^2 + y^2)} [-y \vec{a}_x + x \vec{a}_y] \cos \left[\omega t - \beta\sqrt{x^2 + y^2} + \frac{\pi}{2} \right]$$

5.

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$$

$$\nabla \times \vec{H} = j\omega \vec{D}$$

$$H_\rho = 0$$

$$H_\phi = j \frac{H_0}{\rho} e^{-j\beta\rho}$$

$$H_z = 0$$

$$\therefore \vec{a}_\rho \left[\frac{1}{\rho} \frac{\partial}{\partial \phi} (H_z) - \frac{\partial}{\partial z} (H_\phi) \right]$$

$$+ \vec{a}_\phi \left[\frac{\partial}{\partial z} (H_\rho) - \frac{\partial}{\partial \rho} (H_z) \right]$$

$$+ \vec{a}_z \left[\frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho H_\phi) - \frac{1}{\rho} \frac{\partial}{\partial \phi} (H_\rho) \right] = j\omega \vec{D}$$

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• نذكر ان الاستنتاجات السابقة
للحالات الان السطحية و

الكروية ولكل التخليلات

الحامية على ∇ مثل

$\nabla \cdot \vec{H}$ و $\nabla \times \vec{H}$

• نجد طريقة العملية

$\nabla \times \vec{H}$

من ال Sheet في جزء

الاحداثيات الاسطوانية

$$\begin{aligned}
& \vec{a}_\rho \left[0 - \frac{\partial}{\partial z} \left(j \frac{H_0}{\rho} e^{-j\beta\rho} \right) \right] \\
& + \vec{a}_\phi \left[0 \quad - 0 \right] \\
& + \vec{a}_z \left[\frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho j \frac{H_0}{\rho} e^{-j\beta\rho}) - 0 \right] = j\omega \vec{\tilde{D}} \\
& \vec{a}_z \left[\frac{1}{\rho} \frac{\partial}{\partial \rho} (j H_0 e^{-j\beta\rho}) \right] = j\omega \vec{\tilde{D}} \\
& \vec{a}_z \left[\frac{1}{\rho} j H_0 e^{-j\beta\rho} (-j\beta) \right] = j\omega \vec{\tilde{D}} \\
& \left[\frac{H_0}{\rho} e^{-j\beta\rho} \right] \vec{a}_z = j\omega \vec{\tilde{D}} \\
& \vec{\tilde{D}} = -j \frac{H_0}{\omega\rho} e^{-j\beta\rho} \vec{a}_z
\end{aligned}$$

$$6. \quad \vec{\tilde{D}} = \nabla \cdot \vec{\tilde{D}}$$

$$\vec{\tilde{D}}_\rho = 0$$

$$\vec{\tilde{D}}_\phi = 0$$

$$\vec{\tilde{D}}_z = -j \frac{H_0}{\omega\rho} e^{-j\beta\rho}$$

$$\begin{aligned}
\therefore \nabla \cdot \vec{\tilde{D}} &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho \vec{\tilde{D}}_\rho) \\
&+ \frac{1}{\rho} \frac{\partial}{\partial \phi} (\vec{\tilde{D}}_\phi) \\
&+ \frac{\partial}{\partial z} (\vec{\tilde{D}}_z) = 0
\end{aligned}$$

$$7. \nabla^2 \tilde{H}_+$$

$$\tilde{H}_+ = j \frac{H_0}{\rho} e^{-j\beta\rho}$$

$$\begin{aligned} & \frac{1}{\rho} \frac{\partial}{\partial \rho} \left[\rho \frac{\partial}{\partial \rho} (\tilde{H}_+) \right] \\ & + \frac{1}{\rho^2} \frac{\partial}{\partial \phi} \left[\frac{\partial}{\partial \phi} (\tilde{H}_+) \right] \\ & + \frac{\partial}{\partial z} \left[\frac{\partial}{\partial z} (\tilde{H}_+) \right] = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left[\rho j H_0 \left(\frac{e^{-j\beta\rho} (-j\beta)\rho - e^{-j\beta\rho}}{\rho^2} \right) \right] \\ & = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left[j H_0 \left(-j\beta e^{-j\beta\rho} - \frac{1}{\rho} e^{-j\beta\rho} \right) \right] \\ & = \frac{j H_0}{\rho} \frac{\partial}{\partial \rho} \left[-j\beta e^{-j\beta\rho} - \frac{1}{\rho} e^{-j\beta\rho} \right] \\ & = \frac{j H_0}{\rho} \left[-j\beta e^{-j\beta\rho} (-j\beta) - \frac{e^{-j\beta\rho} (-j\beta)\rho - e^{-j\beta\rho}}{\rho^2} \right] \\ & = \frac{j H_0}{\rho} \left[-\beta^2 e^{-j\beta\rho} + \left(\frac{j\beta e^{-j\beta\rho}}{\rho} \right) + \frac{e^{-j\beta\rho}}{\rho^2} \right] \\ & = \frac{j H_0 e^{-j\beta\rho}}{\rho^3} \left[1 + j\beta - \beta^2 \rho \right] \end{aligned}$$